

EDUGENIE

Google Gemini Powered Learning Assistant

2. Requirement Analysis

Functional Requirements

- Question and Answer: accept learner questions and return AI-generated answers.
- Concept Explanation: explain topics in simplified language.
- Quiz Generation: create three multiple-choice questions with four options each from supplied content.
- Summarization: convert long educational passages into concise revision-friendly summaries.
- Learning Recommendations: generate structured learning paths from beginner to advanced level.

Backend Requirements

- FastAPI framework for RESTful API development.
- Uvicorn ASGI server for local execution.
- Separate Python modules for explanation, Q&A, quiz, summary, and learning-path functionality.
- API endpoints: /qa, /explain, /quiz, /summarize, and /learn/recommendations.

Frontend Requirements

- HTML interface with a task-selection dropdown.
- Text area for user input.
- Submit button for sending requests.
- Result area for displaying generated responses.
- Responsive CSS styling.

AI Requirements

- Google Gemini API key for cloud-based generative AI features.
- A lightweight local model, LaMini-Flan-T5-783M, for concept explanation as specified in the documentation.
- Error handling for invalid or incomplete AI responses.

Environment Requirements

- Python 3.10 or later.

- Internet access for cloud AI API requests.
- A computer capable of running the lightweight web application and its dependencies.

Source basis: EduGenie project documentation provided by the user.